DPR: History and Physical Exam of the Shoulder and Elbow

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College of Osteopathic Medicine

SESSION OBJECTIVES

THE STUDENT PHYSICIANS WILL BE ABLE TO:

1. **Demonstrate** the components of the **history and physical examination** of the elbow and shoulder

2. **Evaluate** the patient utilizing **inspection, palpation, active range of motion, passive range of motion, strength testing** of the elbow and shoulder

3. **Apply** special testing of :the shoulder including **painful arc test, the empty can test, drop-arm test*, Yergason test, Apprehension test,* Hawkin's test, Neer's test and Apley scratch test**

:and the elbow **including tests for medial and lateral epicondylitis, laxity of the elbow and ulnar nerve irritation**

4. **Distinguish** **normal versus abnormal examination** findings.

5. Properly **document** normal and abnormal findings of the shoulder and elbow examination in the objective component of the **S*O*AP note**.

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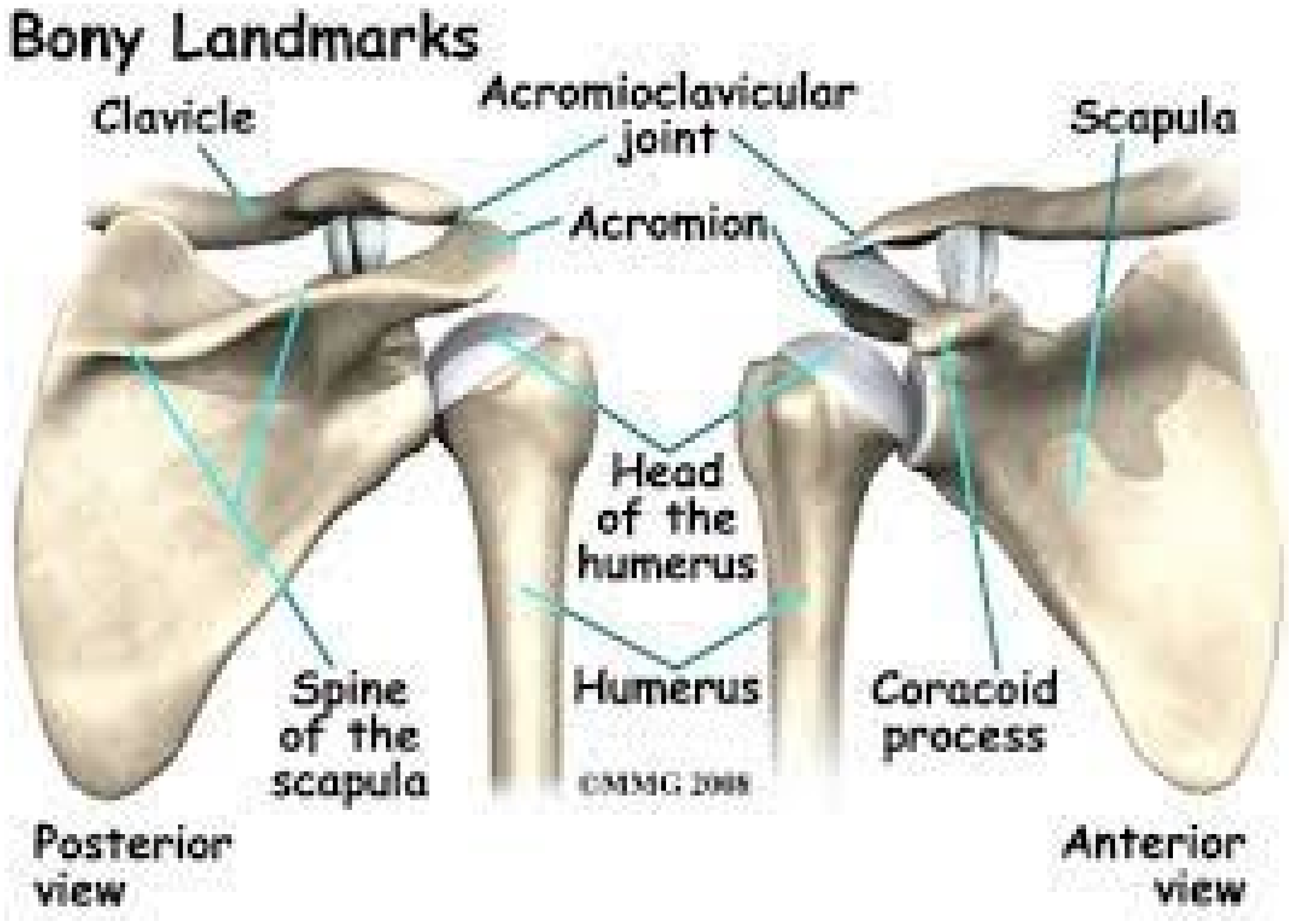
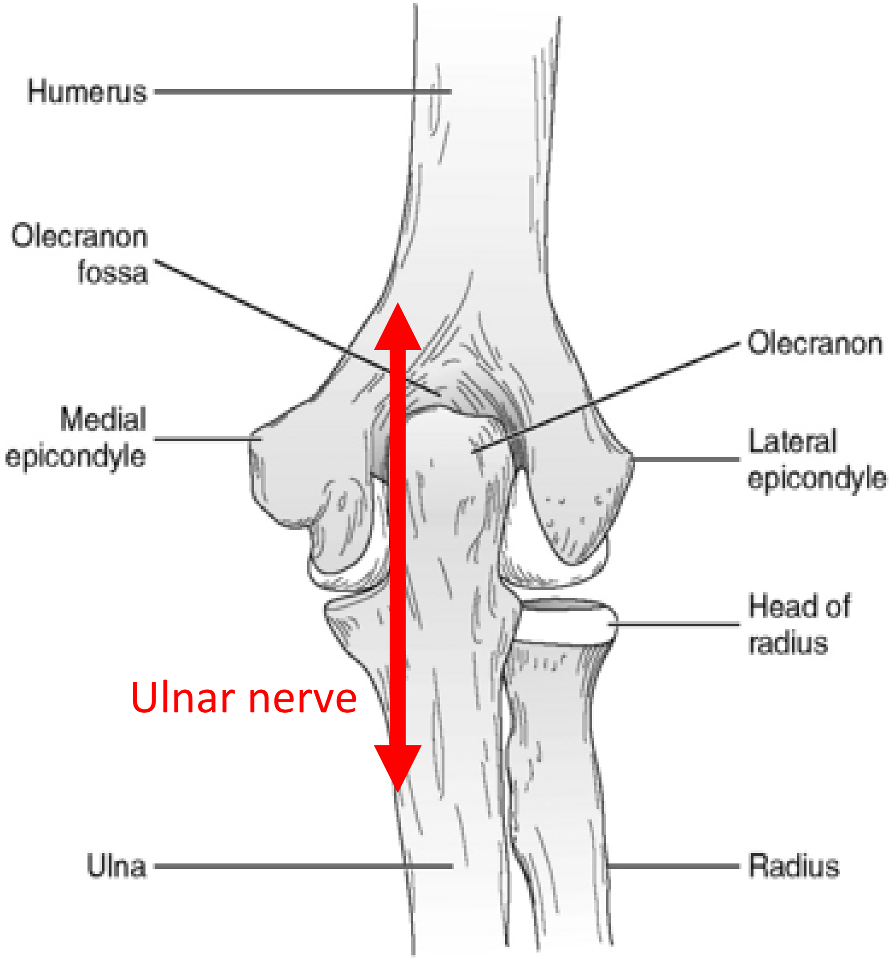
Source: Course Syllabus


Bony Landmarks
Clavicle
Acromioclavicular joint
Scapula
Acromion
Head of the humerus
Humerus
Spine of the scapula
Coracoid process
©MMG 2008
Posterior view
Anterior view

Elbow Anatomy

- Olecranon
- Epicondyles
- Synovium (between epicondyles and olecranon)
- Ulnar nerve



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Clinical Decision Making

- How is MSK history different than all the rest?
  - Prior injuries?
  - Repetitive movements at home or work/hobbies and habits
  - Patient may not know the etiology

- After asking your history questions…
  - Time to think about what your potential diagnosis list will look like
- How will this list look after you perform your physical exam?
  - Will things be ruled in or excluded?
  - What other data will you need from the exam and special tests?

MSK History-Where

- Articular or Extraarticular
- Point with one finger
- Deep or superficial
- Pinpoint or diffuse
- Mono, Oligo or Polyarticular
- Is there migration from joint to joint
- Intermittent or constant
- Symmetrical or asymmetrical
- Does the pain radiate (hip to groin)

MSK History Onset/Progression

- Super important!!!!
- When did the pain start
- Was there an event that precipitated the pain(need details)
- Does it come and go OR has it come and is worsening
- How long does the pain last
- How often do you have the pain
- Does it vary over the course of the day
- Is it worse with rest or with activity

MSK History Provocation/Remitting factors

- Does anything worsen the pain
- Does anything make the pain better
- Specifically ask about movement, rest, medication, ice, heat
- Do the alleviating factors take the pain to zero or only offer a bit of relief

MSK History Quality

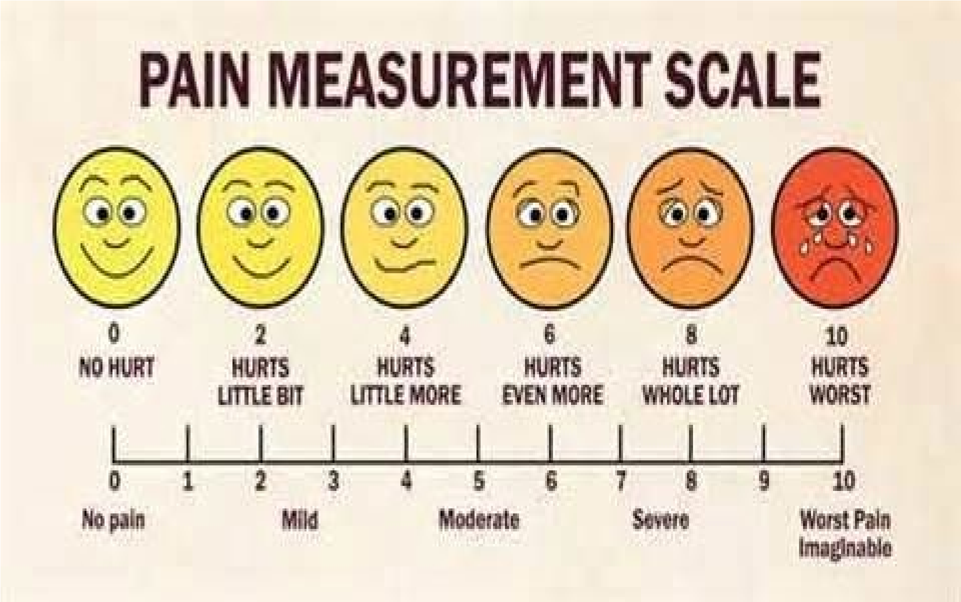
- Describe the pain. Initially don't offer options. It is important to allow the patient to use their own words.
- If the patient is having difficulty describing the pain, then you can offer suggestions.
- The quality of the pain in MSK is not the most helpful information but should still be gathered.

MSK History Radiation

- Pain can radiate up and down
- The longer the pain has been present, the further it can radiate
- When assessing chronic pain (more than 3 months old) don’t get fooled by the radiating pain

MSK History Severity

- Dig a little deeper with MSK
- Initial severity, current severity, worst severity, usual severity
- VAS: visual analog scale is based on the patients report of pain. We most often use a scale of 1-10, but what does that mean? Is your 5 =to my 5?



ASSOCIATED SYMPTOMS-LOCAL

- Is there local swelling, heat, redness,
- Is there numbness, pins/needles, local weakness
- Is there a rash, a bruise, a hematoma
- Is there a deformity, a lump, a bump, a lesion
- Is there stiffness, limited movement, painful movement
- Does it hurt only when they move the joint, or even when someone else moves it
- Has activity been hampered by the local symptoms
- Are ADLs being affected: hair combing, brushing teeth, getting dressed

ASSOCIATED SYMPTOMS- CONSTITUTIONAL

- Fever, chills, night sweats
- Fatigue
- Weight loss,
- Rash
- Generalized weakness

Physical Exam MSK

Always look before you touch

ALWAYS EXAMINE BOTH SIDES

Always touch before you move

- ALWAYS EXAMINE BOTH SIDES

Ask the patient to move first (AROM)and then you can move the patient (PROM)

NEXT……Perform strength testing

ALWAYS EXAMINE BOTH SIDES

- ALWAYS EXAMINE BOTH SIDES
- And finally perform special testing

The MSK Exam always follows the same format

Inspection

Palpation

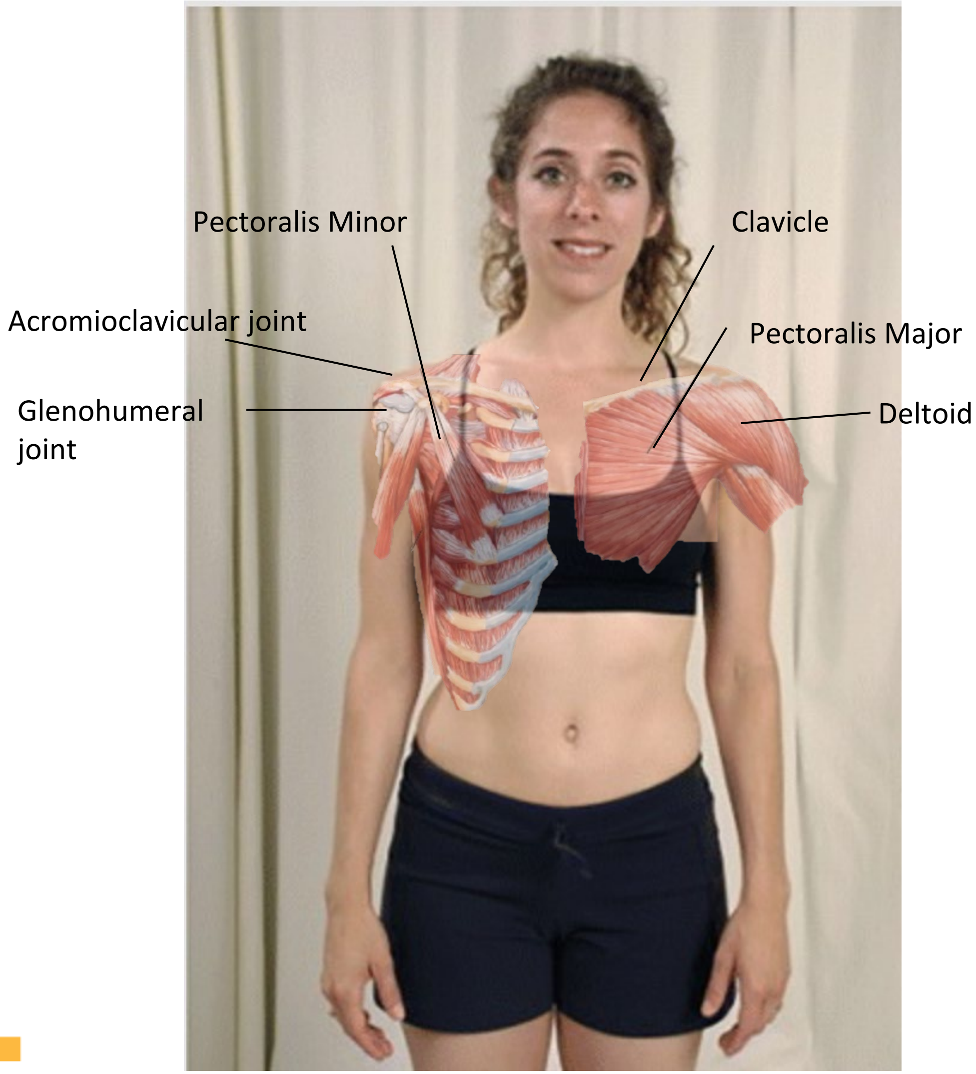
ROM – Active then passive

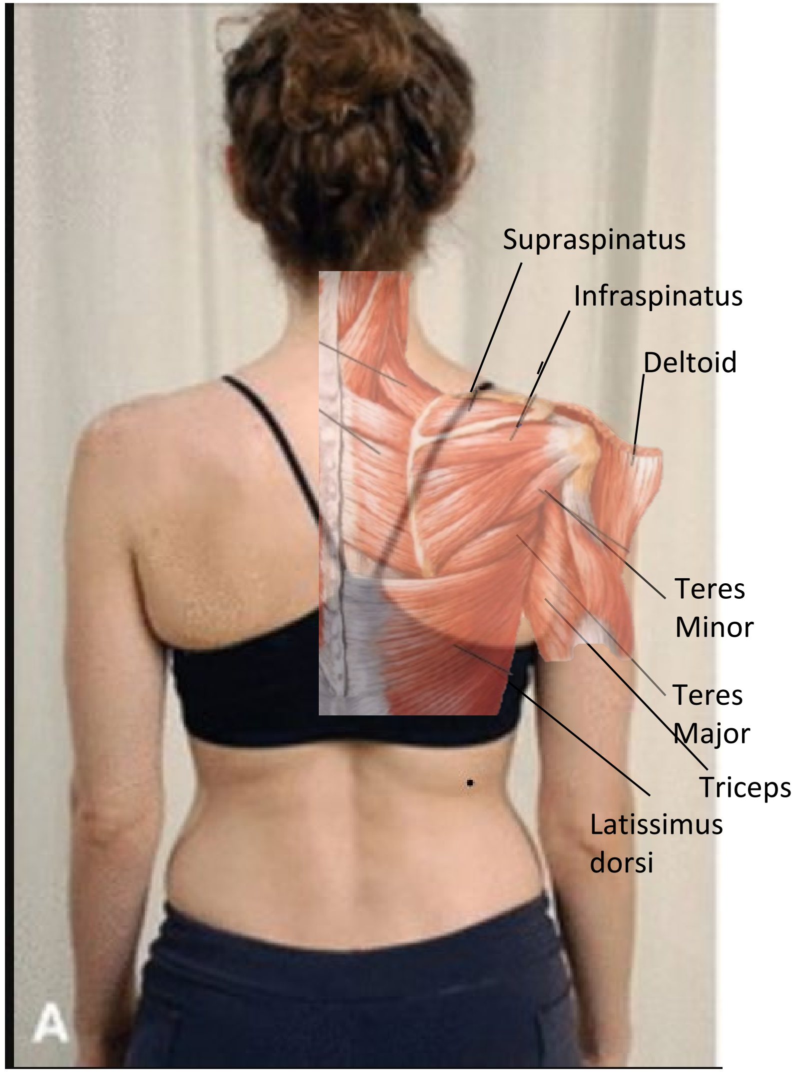
MMT

Special testing

Check sensory if there is a neuro component to the patient's complaint

Inspection of Shoulder



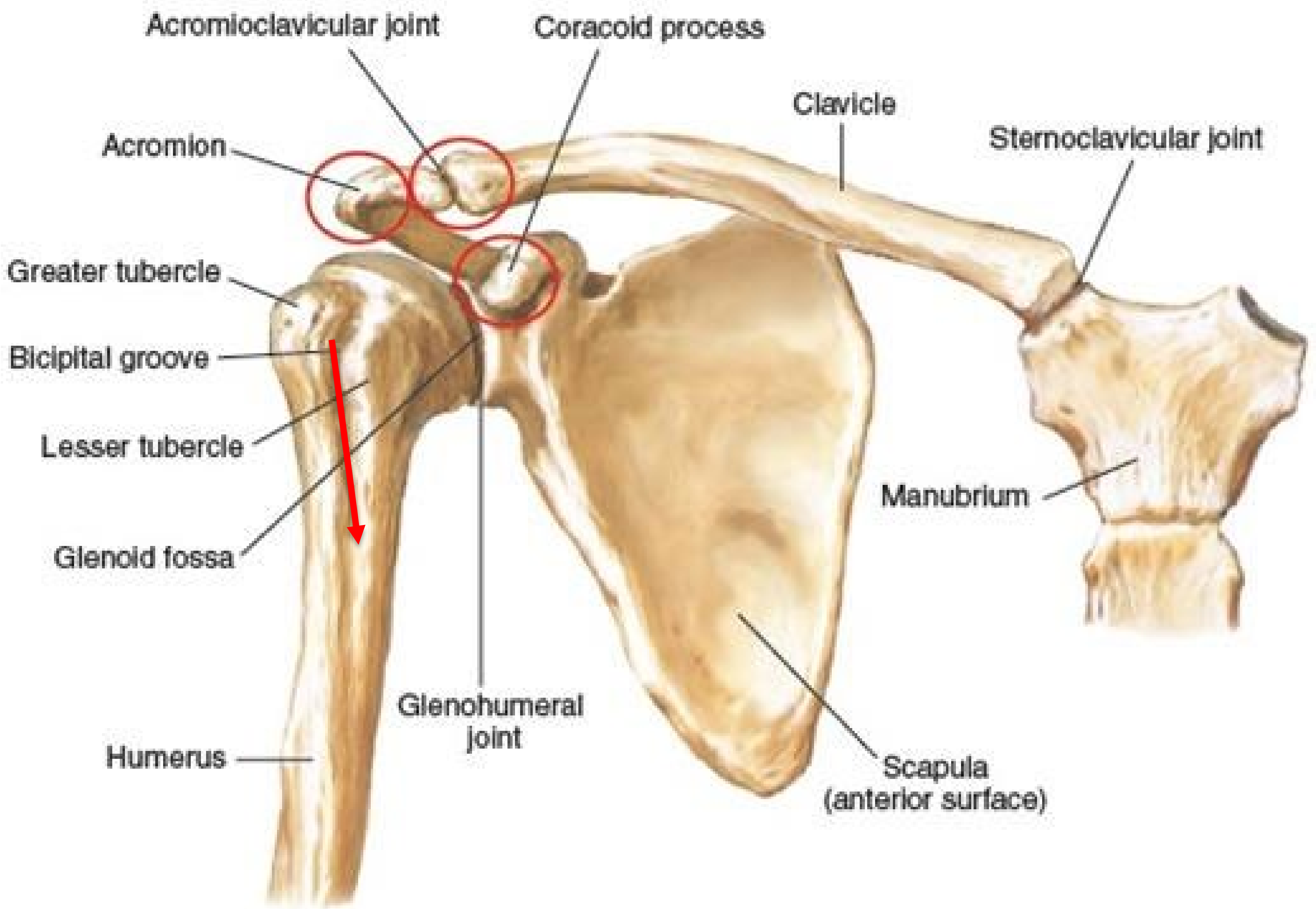


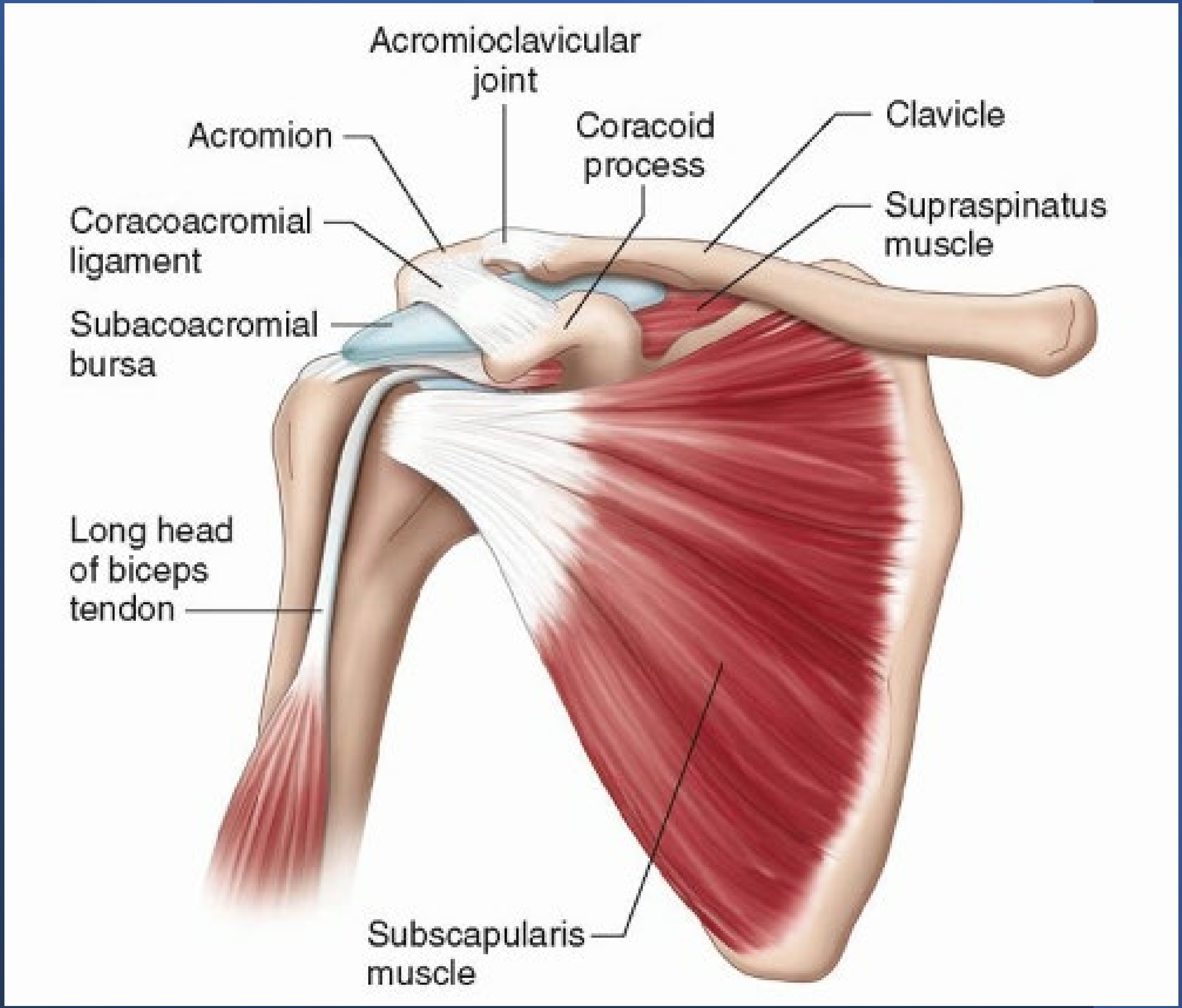
- INSPECT FROM BEHIND AND IN FRONT
- Look for asymmetry, atrophy, scars, rashes, evidence of injury or inflammation

Source: VH Dissecter Examination of the shoulder, Part 1

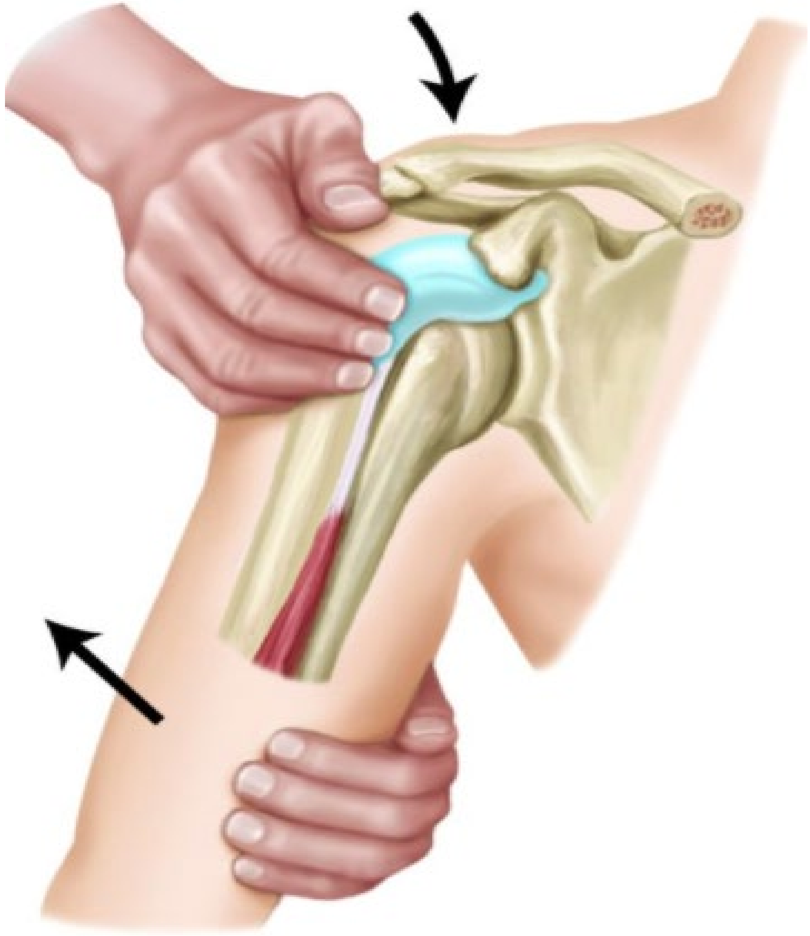
Palpation of the Shoulder

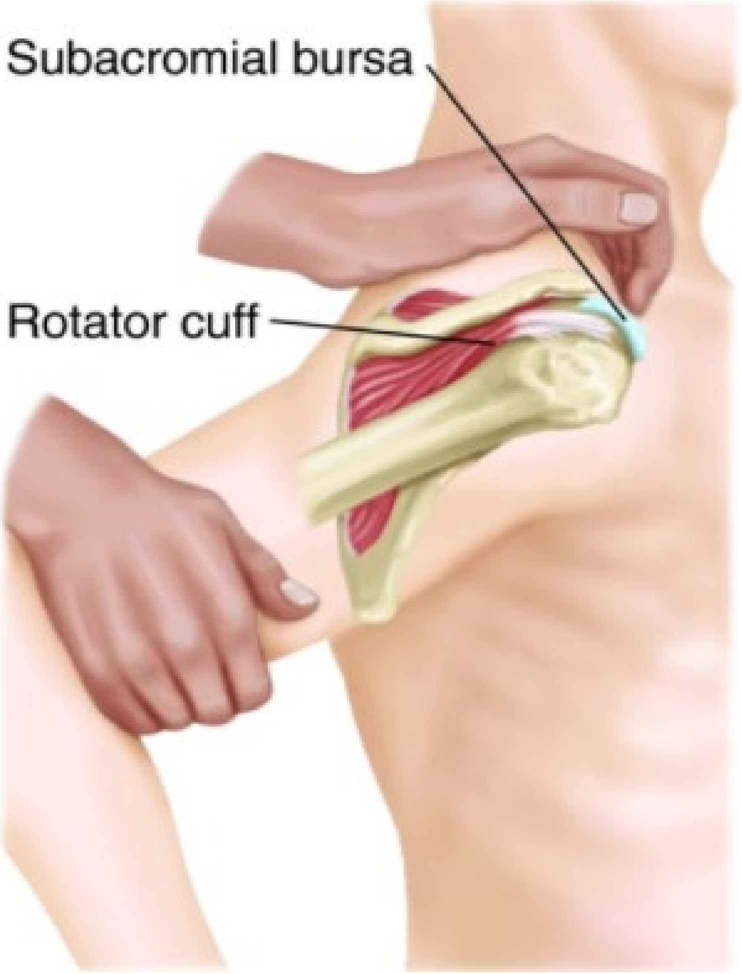
- Bony contours
- Areas of pain
- SC joint
- AC joint
- Clavicle
- Acromion
- Spine of the scapula
- Coracoid process
- Greater tubercle
- Biceps tendon
- Subacromial bursa


Acromioclavicular joint
Coracoid process
Clavicle
Sternoclavicular joint
Acromion
Greater tubercle
Bicipital groove
Lesser tubercle
Glenoid fossa
Manubrium
Glenohumeral joint
Humerus
Scapula (anterior surface)


Acromioclavicular joint
Acromion
Coracoid process
Clavicle
Coracoacromial ligament
Supraspinatus muscle
Subacoacromial bursa
Long head of biceps tendon
Subscapularis muscle

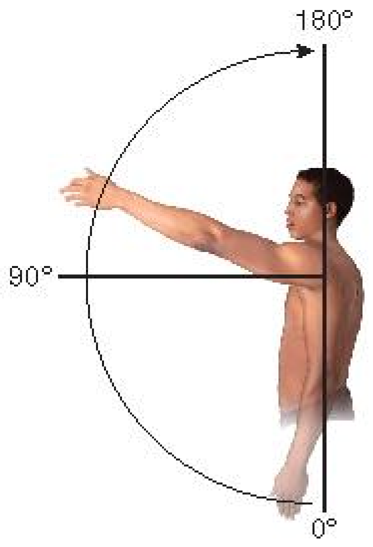
Palpation

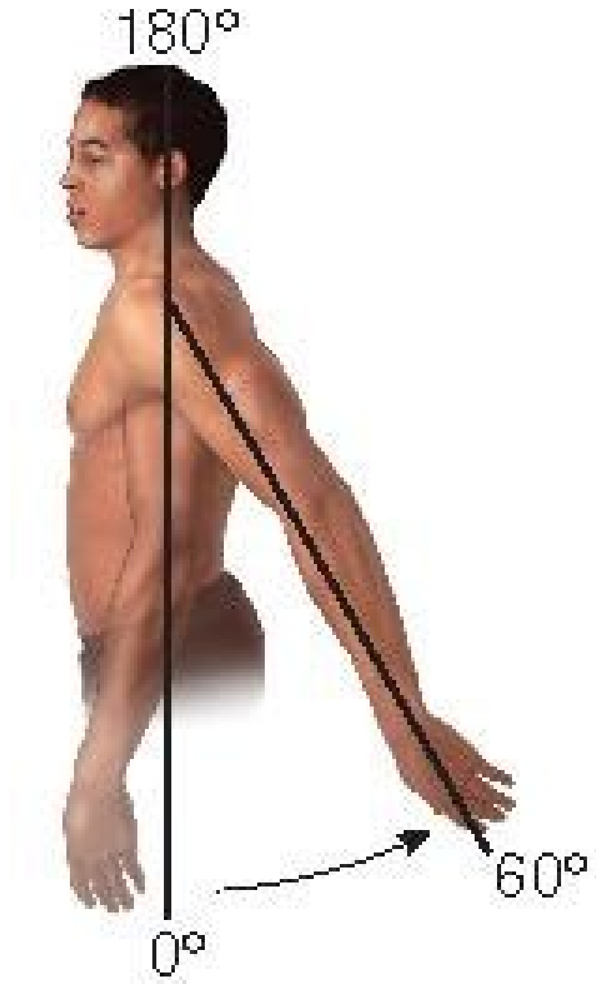




AROM FLEXION

- **Principal Muscles Affecting Movement**
  Anterior deltoid, pectoralis major (clavicular head), coracobrachialis, biceps brachii
  Patient Instructions
  "Raise your arms in front of you and overhead."





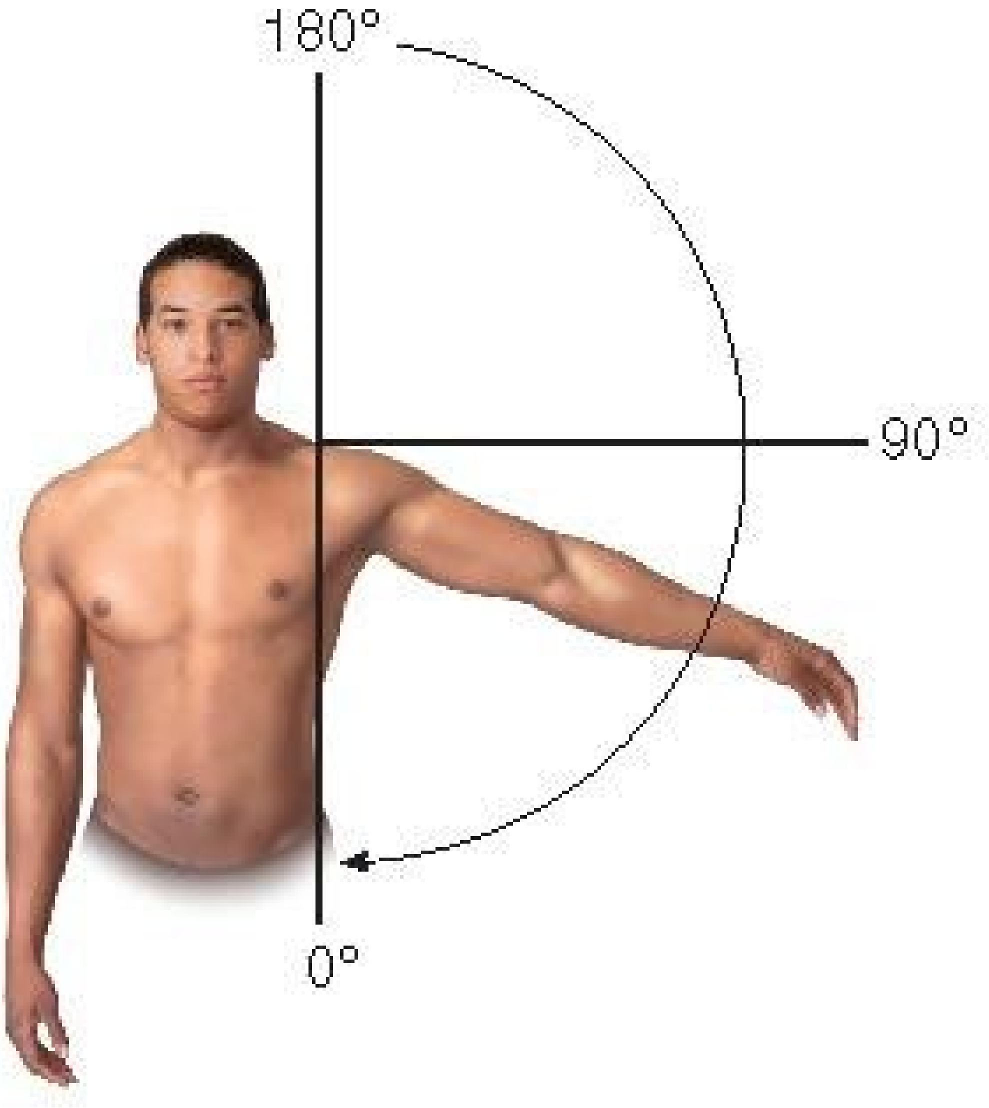
Extension

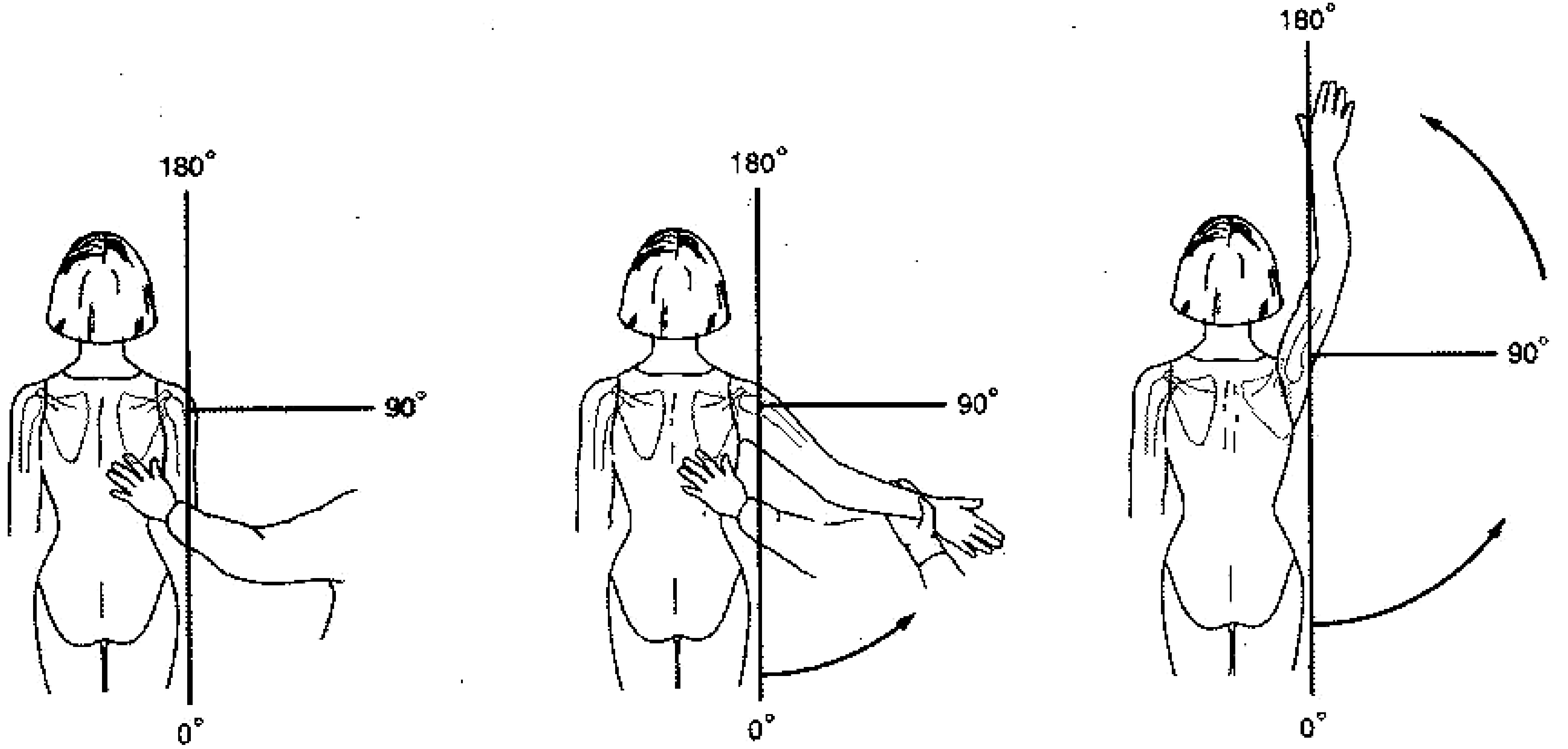
- **Extension**

Principal Muscles Affecting Movement
Latissimus dorsi, teres major, posterior deltoid, triceps brachii (long head)
Patient Instructions
“Raise your arms behind you.”

•Adduction

Principal Muscles Affecting Movement
Pectoralis major, coracobrachialis, latissimus dorsi, teres major, subscapularis
Patient Instructions
"Cross your arm in front of your body."





First 10-15 degrees: SS & IS
Then deltoid takes over

•Internal Rotation

Principal Muscles Affecting Movement
Subscapularis, anterior deltoid, pectoralis major, teres major, latissimus dorsi
Patient Instructions
“Place one hand behind your back and try to touch your shoulder blade.”
Identify the highest midline spinous process the patient is able to reach.

•External Rotation

Principal Muscles Affecting Movement
Infraspinatus, teres minor, posterior deltoid, supraspinatus (especially with arm overhead)
Patient Instructions
“Raise your arm to shoulder level; bend your elbow and rotate your forearm toward the ceiling.”
OR
“Place one hand behind your neck or head as if you are brushing your hair.”

Motor Manual Testing

- MMT should then be performed in all available ranges of motion
- Graded on a scale of 0-5
- 0 (zero) = no visible muscle movement at all
- 1 (trace) = muscle contraction felt but no visible movement of muscle
- 2 (poor) = movement through complete ROM with gravity eliminated
- 3 (fair) = holds test position with no added pressure
- 4 (good) = holds test position against moderate pressure
- 5 (normal) = holds test position against strong pressure

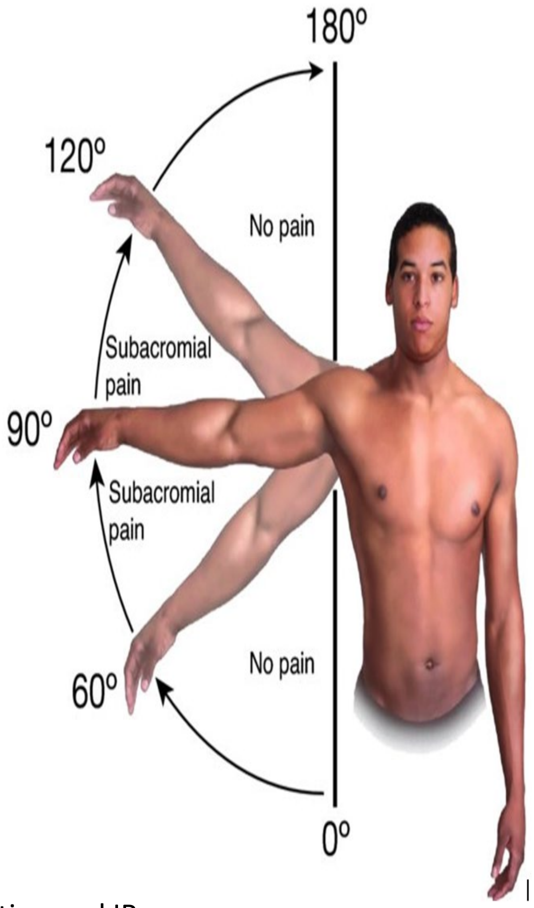
- Crossover or crossed body adduction test - *AC joint*

Apley scratch test – *Shoulder rotation*

Tests abduction and external rotation.

Tests ADDuction and IR

Painful arc test- *rotator cuff tendinitis*



Special Tests

Neer's impingement sign- *impingement*

Neer is close to the ear

Forced flexion

Hawkin's impingement sign- *impingement*

Arm looks like a hawk wing

Forced internal rotation

Empty can test-
Supraspinatus tear

Patient is pushing up while you are pushing down

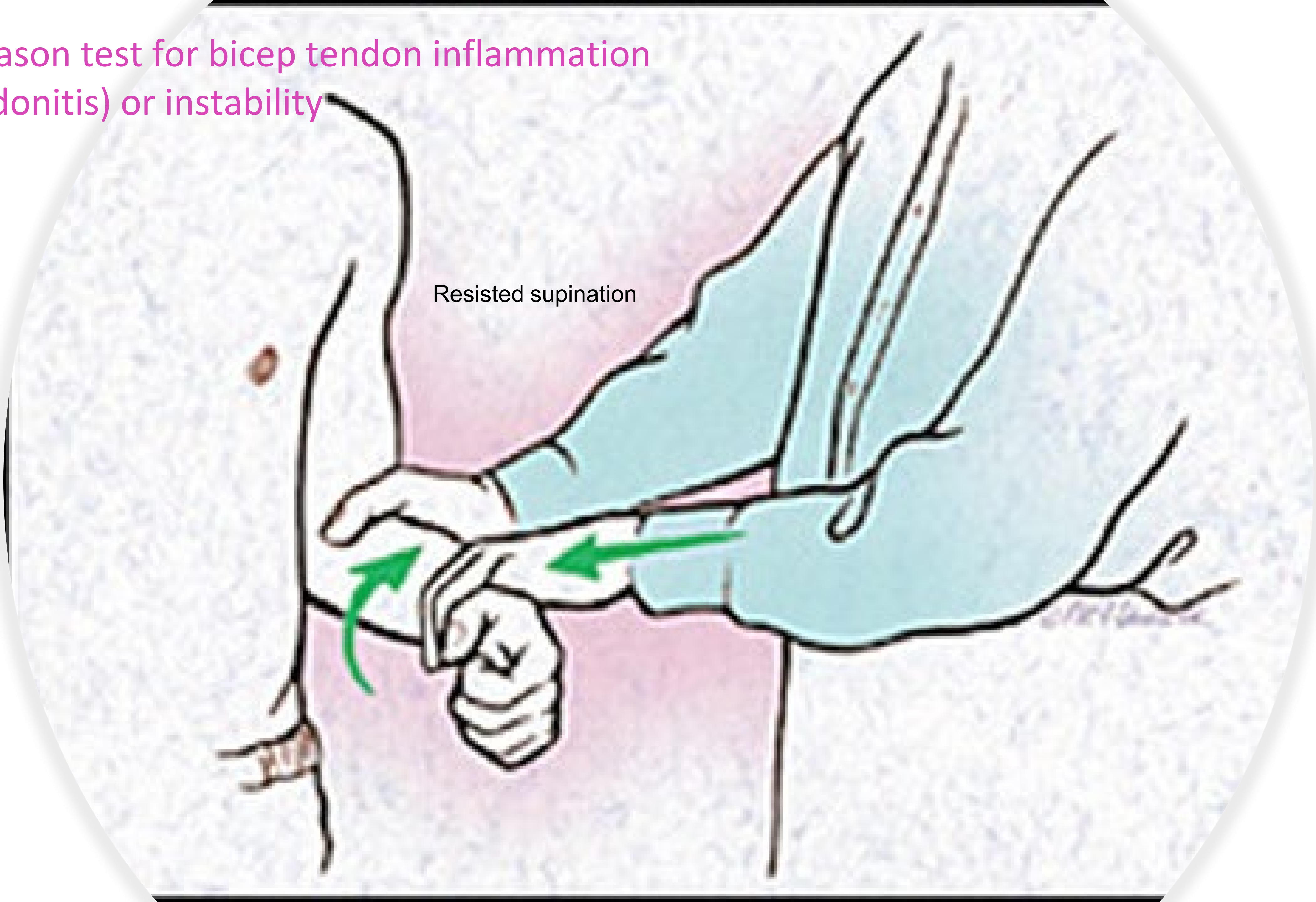
Rotator cuff instability:

Drop arm test (Codman's sign) ✪

- Active
- Seated
- Abduct the arm to 90°, lower the arm slowly
- (+): Not able to lower the arm slowly or drops suddenly = rotator cuff tear (usually supraspinatus)

Apprenhension test for anterior instability for dislocation

Yergason test for bicep tendon inflammation (tendonitis) or instability



Bicep Tendonitis

- anterior shoulder pain
- pain is exacerbated by lifting objects and overhead activities
- improves with rest
- localized tenderness over the bicipital groove
- painful with resisted elbow flexion and forearm supination(Yergason)

- Pearls:
  - Cervical pathology may masquerade as shoulder pain(C5)
  - Throwing or racquet sport ask phase (cocking, release, follow through)
  - Posterior dislocation – arm is usually in internal rotation at time of injury
  - Anterior dislocation – arm is usually in external rotation at time of injury

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Elbow Inspection

- Similar to shoulder:
- Asymmetry, lumps, bumps, scars, bruises, rashes, signs of inflammation or injury

Olecranon bursitis

Elbow Palpation

- Olecranon
- Epicondyles
- Synovium (between epicondyles and olecranon)
- Ulnar nerve

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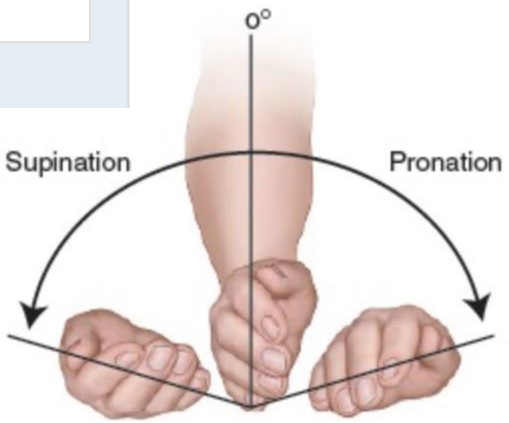
Elbow Range of Motion

Elbow Range of Motion

Elbow Movement	Primary Muscles Affecting Movement	Patient Instructions
Flexion	Biceps brachii, brachialis, brachioradialis	*"Bend your elbow."*
Extension	Triceps brachii, anconeus	*"Straighten your elbow."*
Supination	Biceps brachii, supinator	*"Turn your palms up, as if carrying a bowl of soup."*
Pronation	Pronator teres, pronator quadratus	*"Turn your palms down."*



Strength Testing

Elbow Flexion

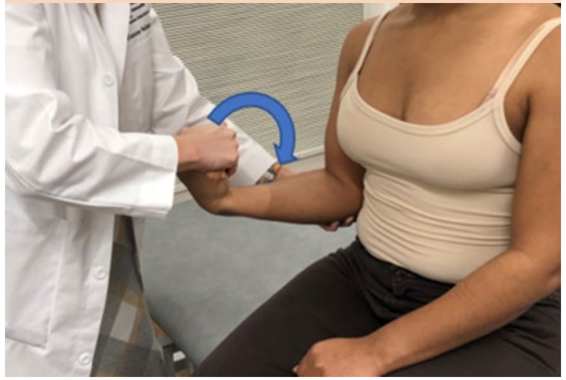
Elbow Extension

Special Tests

Tinel's of Elbow-testing for cubital tunnel syndrome

Cozen Test-Lateral epicondylitis testing

Medial epicondylitis testing



Elbow

- Pearls
  - Lateral epicondyles (LES)
    - Extensors
    - Supinators
  - Medical epicondyles (MFP)
    - Flexors
    - Pronators
- Do not miss
  - Tendon rupture
  - Dislocation
  - Fractures

Source: Stockard, A. (2010). *Osteopathic clinical joint exam*. Madison, Wisconsin: American Osteopathic Academy of Sports Medicine.

Shoulder pain

Common	Less common	Not to be missed causes outside the shoulder girdle
Rotator cuff pathology (strain, tear, tendinopathy)	Other muscle tear (pectoralis major, long head of biceps (LHB))	Somatic referred pain (cervical and thoracic spine, myofascial structures)
Glenohumeral instability (traumatic, atraumatic, congenital)	Adhesive capsulitis	Tumour (bone tumours in proximal humerus)
Biceps related pathology (SLAP labral tears, biceps tendinopathy, tenosynovitis)	Neurovascular entrapment or traction injury (suprascapular nerve, long thoracic nerve, axillary nerve, axillary vein compression, thoracic outlet syndrome)	Visceral referred pain (e.g. diaphragm, gall bladder, heart, spleen, apex of lungs)
Posterior shoulder stiffness (GIRD)	Fractures (scapula, humerus, stress fracture of coracoid process)	
Scapular pathology and dysfunction, including AC and SC joint pathology (sprain, fracture, dislocation, clavicular fracture)	Snapping scapula	

SLAP = superior labrum anterior to posterior; GIRD = glenohumeral internal rotation deficit; AC = acromioclavicular; SC = sternoclavicular

Elbow Pain

Table 1. Differential Diagnosis of Elbow Pain Based on Anatomic Location

Anterior
Anterior capsule strain
Biceps tendinopathy
Gout
Intra-articular loose body
Osteoarthritis
Pronator syndrome
Rheumatoid arthritis

Lateral
Lateral epicondylitis
Osteochondral defect
Plica
Posterolateral rotatory instability
Radial tunnel syndrome/posterior interosseous nerve syndrome

Medial
Cubital tunnel syndrome
Medial epicondylitis
Ulnar collateral ligament injury
Valgus extension overload syndrome

Posterior
Olecranon bursitis
Olecranon stress fracture
Osteoarthritis
Posterior impingement
Triceps tendinopathy

Kane, SF, Lynch, JG, Taylor, JC. Evaluation of elbow pain in adults. *AAFP. 2014; 89(8):649-657*

How to document the MSK exam

- Which **joint** are you documenting?: you cannot say normal ROM in right upper extremity unless you measured every joint in that arm
- Which **side** are you documenting?: right or left or both
- ROM should be documented as: Patient's finding/normal ROM
- MMT is on a scale from 0-5
- **EX: On examination of the left shoulder Flexion 120/180, abduction 110/180, IR 45/90, ER 70/90, Adduction and extension were normal**
- **MMT revealed 5/5 strength in adduction, extension and 4/5 strength in abduction, flexion, IR and ER.**
- **The right shoulder exam revealed normal ROM and 5/5 strength in all the above movements**

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https://exchange.scholarrx.com/brick/muscles-of-the-upper-extremity